IMPACT ASSESSMENT

Kampung Makmur Sdn Bhd

● Ready for IC

KM

agriculture · Malaysia · Generated 2026-11-10 · IRIS+ 5.3c · Source: kampung_makmur_pig_farm.pdf

Confidential — prepared for internal investment and LP use.

● Extracted by rules ● 3/22 scores estimated

Recommendation

Ready for IC

Expected impact is quantified and the evidence is good enough to take to the investment committee.

- evidence quality good (63/100)
- people impact P50 61 depth-weighted person-years
- climate impact P50 859 tCO2e

Ready, with 18% of the core metric set reported.

Evidence quality: 63/100 · good

Expected impact	Evidence quality	Top SDG	Greenwashing risk
61 depth-weighted person-years · P10 26 – P90 140	**63**/100 good · 18% core metrics	7 **SDG 7** Affordable and Clean Energy · 43/100	**38**/100 Substantive with gaps · finding threshold 60

What would change our mind

The evidence that would narrow the range most, in order.

1. **Report the sector's core metrics: Payments to Supplier Individuals: Smallholder, Water Withdrawn, Waste Produced: Total.**
   Only 18% of the core set is reported.

Expected impact

How much change, for whom, and how sure we are. Reach × depth × duration, less what would have happened anyway, credited to the company and weighted by the

chance it happens. Ranges come from the evidence behind each figure.

Evidence quality

63/100

good · NESTA 2/3 · third-party verified

Data completeness

18%

of the sector's core metrics reported

People · households

61 depth-weighted person-years



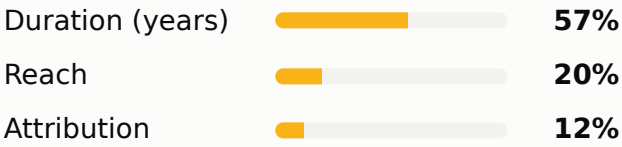
P10 26 P50 61 P90 140

moderate uncertainty: P90 is ×5.5 the P10

What goes into it

Reach	120	120 households reported (NESTA 3)
Depth of change	31%	31% change reported (NESTA 3)
Duration (years)	3	typical for agriculture
Without deadweight	100%	netted out by the comparison group
Attribution	70%	share credited to the enterprise
Probability of success	80%	evidence level 3

What drives the uncertainty



> “A 2024 randomized pilot with Universiti Putra Malaysia compared 120 outgrower households to a matched control group and found 23% higher household food-security scores and 17% higher child school attendance.”●●●○○

Climate

860 tCO2e



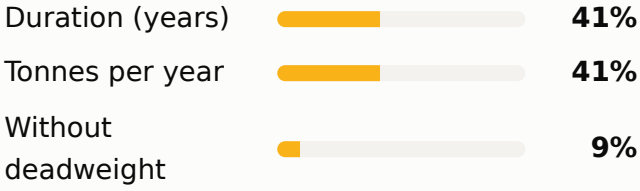
P10 300 P50 860 P90 2,300

wide uncertainty: P90 is ×7.5 the P10

What goes into it

Tonnes per year	920	920 tCO2e reported (NESTA 2)
Duration (years)	3	typical for agriculture
Without deadweight	70%	no comparison group: ~30% would have happened anyway
Attribution	70%	share credited to the enterprise
Probability of success	65%	evidence level 2

What drives the uncertainty



> “Our biogas plant captures methane from manure lagoons and generates 1.8 GWh of electricity per year, replacing ~920 tonnes CO2e of grid emissions and cutting our grid electricity purchases by 74%.”●●○○○

Targets stated (to compare with actual results later)

- Over the next five years we target: 45% reduction in Scope 1+2 emissions intensity (kg CO2e / kg live weight); 80% of breeding sows in group-housing >= 10 weeks after weaning; 35 outgrower households onboarded; and a SBTi near-term target submission by 2027.

Impact at a glance

Material SDGs and how the company's activities lead to outcomes, from the claims in

the documents.



- SDG 7 · Affordable and Clean Energy — 43
- SDG 2 · Zero Hunger — 39
- SDG 13 · Climate Action — 37
- SDG 6 · Clean Water and Sanitation — 30
- SDG 5 · Gender Equality — 28
- SDG 12 · Responsible Consumption and Production — 28

WHAT THEY DO

Kampung Makmur is a 3,000-sow commercial pig farm in Johor, Malaysia, operating an integrated breed-to-finish facility with a licensed biogas-to-power plant.

OUTPUTS

We are SEDEX-audited and have a 4-tier grievance mechanism.

Risks we actively manage include: (i) ASF biosecurity - all farms are fenced, SPF breeding stock, shower-in/out…

We report against IRIS+, SDG 2/6/7/8/12/13, GRI standards, and have commissioned a first independent ESG audit for…

OUTCOMES

Water withdrawal per kg of live weight dropped 31% versus our 2020 baseline, independently verified by SIRIM.

A 2024 randomized pilot with Universiti Putra Malaysia compared 120 outgrower households to a matched control group and…

Since 2022 we have reduced antibiotic use by 62% through vaccination and group-housing, and replaced 100% of…

TARGETS

Over the next five years we target: 45% reduction in Scope 1+2 emissions intensity (kg CO2e / kg live weight); 80% of…

Five Dimensions of Impact

Scored 1–5. The tick shows the sector benchmark where one exists.

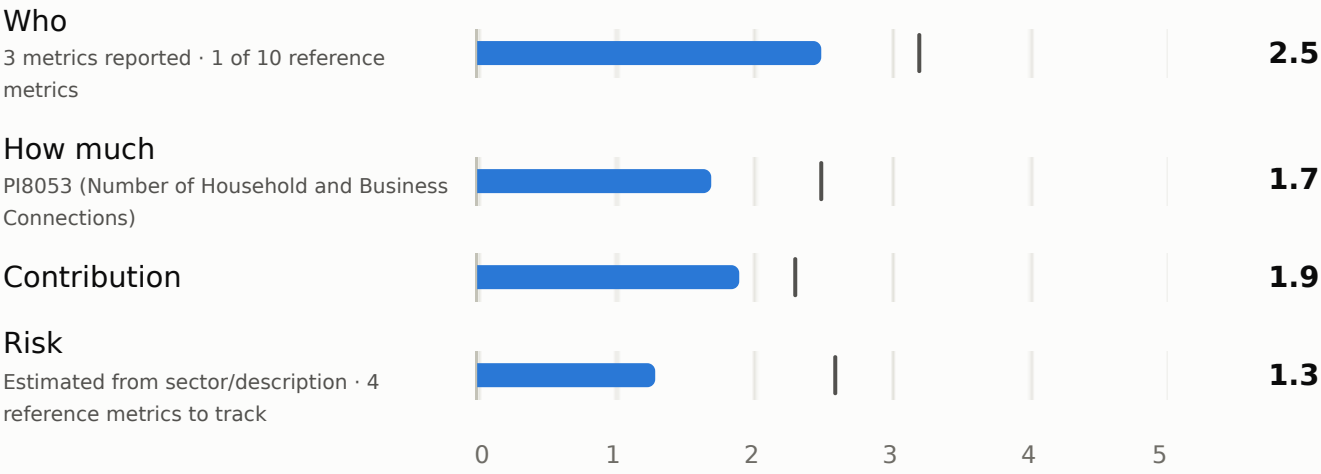
partial evidence

What

2 metrics reported · 1 of 25 reference metrics



2.3



Contribution: enterprise and investor

Enterprise contribution

75/100

Evidence level 3 · compared with a comparison group

“A 2024 randomized pilot with Universiti Putra Malaysia compared 120 outgrower households to a matched control group and found 23% higher household food-secur...”

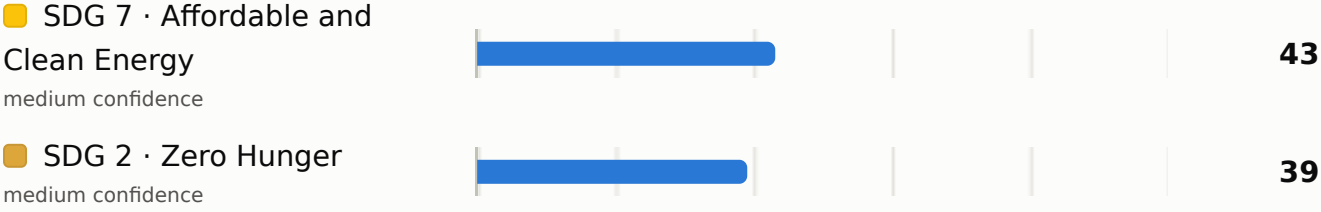
Investor contribution

0/100

Not described in these documents. Add the fund's role from the investment memo.

SDG alignment

Material goals only (claimed, inferred from the business, or evidenced by a goal-specific metric), scored 0–100.





Other goals (context only): 3, 9, 10, 11, 14, 16, 17

Evidence ledger

Every claim found in the documents and every metric used in scoring.

Claims (10)

Claim	Type	NESTA level	Evidence	Metric
Since 2022 we have reduced antibiotic use by 62% through vaccination and group-housing, and replaced 100% of conventional slatted housing with deep-litter, enrichment-enabled pens benchmarked against the EU Welfare Quality protocol. p. 3	outcome	●●○○○	—	—
Our biogas plant captures methane from manure lagoons and generates 1.8 GWh of electricity per year, replacing ~920 tonnes CO2e of grid emissions and cutting our grid electricity purchases by 74%. p. 4	outcome	●●○○○	—	**OI2496** Energy Generated for Use: Renewable **OI2764** Greenhouse Gas Emissions Avoided

Claim	Type	NESTA level	Evidence	Metric
Water withdrawal per kg of live weight dropped 31% versus our 2020 baseline, independently verified by SIRIM. p. 4	outcome	●●●○○	✓ Third-party verified ✓ Baseline comparison	
We employ 142 staff, of whom 38% are women and 64% are from Kluang district; median wage is 42% above the Johor agricultural minimum and every worker is enrolled in SOCSO and EPF. p. 5	outcome	●●○○○	—	**OI8869** Permanent Employees: Total **OI6213** Full-time Employees: Female
We run an outgrower programme with 18 smallholder piggeries, supplying feed and veterinary support at wholesale cost; participating smallholders report average income uplifts of RM 1,850 per month (baseline: RM 900). p. 5	outcome	●●○○○	✓ Baseline comparison	**PI9991** Supplier Individuals: Smallholder
A 2024 randomized pilot with Universiti Putra Malaysia compared 120 outgrower households to a matched control group and found 23% higher household food-security scores and 17% higher child school attendance. p. 5	outcome	●●●○○	✓ Controlled evaluation	
Over the next five years we target: 45% reduction in Scope 1+2 emissions intensity (kg CO2e / kg live weight); 80% of breeding sows in group-housing >= 10 weeks after weaning; 35 outgrower households onboarded; and a SBTi near-term target submission by 2027. p. 5	intent	●●○○○	—	—

Claim	Type	NESTA level	Evidence	Metric
We are SEDEX-audited and have a 4-tier grievance mechanism. p. 5	output	●●○○○	✓ Audited	—
Risks we actively manage include: (i) ASF biosecurity - all farms are fenced, SPF breeding stock, shower-in/out protocol, quarterly third-party audit by the Department of Veterinary Services; (ii) nutrient run-off during monsoon - on-site retention ponds sized to 1-in-100-year storm, automated ammonia sensors; (iii) community odour - a covered-lagoon retrofit was completed in Q3 2025 and the formal complaints rate has dropped from 14 to 2 per year. p. 6	output	●●○○○	✓ Audited	—
We report against IRIS+, SDG 2/6/7/8/12/13, GRI standards, and have commissioned a first independent ESG audit for FY2026. p. 7	output	●●○○○	✓ Audited	—

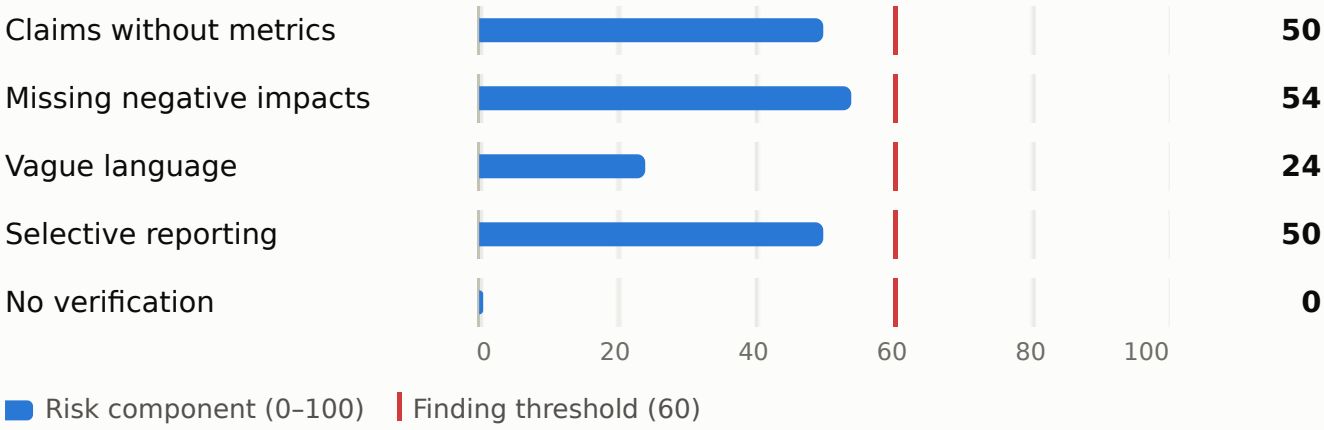
Metrics (5)

Metric	Value	Origin
OI2496 · Energy Generated for Use: Renewable	1,800 MWh	extracted from text
OI2764 · Greenhouse Gas Emissions Avoided	920 tCO2e	extracted from text
OI6213 · Full-time Employees: Female	54 (derived: 38% of 142)	derived
OI8869 · Permanent Employees: Total	142	extracted from text

Metric	Value	Origin
PI9991 · Supplier Individuals: Smallholder	18	extracted from text

Greenwashing review — 38/100 · Substantive with gaps

Components of the 0–100 risk score. At 60 or above risk is a finding; below it mostly reflects missing data.



Risks and opportunities

Potential negative impacts

Harms a business like this can cause, scored severity × likelihood (ESRS impact materiality). A control described in the documents lowers likelihood. Unaddressed material items are questions for diligence, not findings.

Negative impact	Severity	Likelihood	Materiality	Control in the documents
Land conversion or deforestation in the supply base ecosystems · PAI 7 (biodiversity-sensitive areas)	3/3	1/3	3/9 · moderate	Not described — ask

Negative impact	Severity	Likelihood	Materiality	Control in the documents
Antibiotic overuse drives antimicrobial resistance public health	3/3	1/3	3/9 · moderate	Described “Founded in 2014 by a second-generation farmer, the company has repositioned itself around animal welfare, antibiotic-stewardship and manure-to-energy circula...” “Founded in 2014 by a second-generation farmer, the company has repositioned itself around animal welfare, antibiotic-stewardship and manure-to-energy circula...”
Manure, run-off or pesticides pollute soil and water communities, ecosystems	3/3	1/3	3/9 · moderate	Described “Risks we actively manage include: (i) ASF biosecurity - all farms are fenced, SPF breeding stock, shower-in/out protocol, quarterly third-party audit by the...” “Our biogas plant captures methane from manure lagoons and generates 1.8 GWh of electricity per year, replacing ~920 tonnes CO2e of grid emissions and cutting...”
Water abstraction stresses local sources communities, ecosystems · PAI 6 / 8 (water)	2/3	1/3	2/9 · low	Described “Water withdrawal per kg of live weight dropped 31% versus our 2020 baseline, independently verified by SIRIM.”

Disclosed in the documents

- ASF biosecurity - all farms are fenced, SPF breeding stock, shower-in/out protocol,

Opportunities

- Food security: improving access to nutritious food for underserved

quarterly third-party audit by the Department of Veterinary Services
- nutrient run-off during monsoon - on-site retention ponds sized to 1-in-100-year storm, automated ammonia sensors
- community odour - a covered-lagoon retrofit was completed in Q3 2025 and the formal complaints rate has dropped from 14 to 2 per year
- Founded in 2014 by a second-generation farmer, the company has repositioned itself around animal welfare, antibiotic-stewardship and manure-to-energy circularity following the regional ASF outbreaks.

Typical for the sector — check in diligence

- Environmental degradation: soil depletion, water pollution from fertilizers/pesticides
- Climate vulnerability: sensitivity to extreme weather events and climate change
- Labor conditions: risk of exploitative labor practices
- Land use change: deforestation or biodiversity loss from land conversion
- Market volatility: commodity price fluctuations affecting farmer incomes
- Environmental pollution: waste management, methane emissions, water contamination

populations
- Poverty reduction: increasing income for smallholder farmers
- Sustainable supply chains: promoting responsible sourcing and production
- Climate adaptation: developing climate-resilient farming practices
- Rural employment: creating jobs in farming communities
- Food security: providing protein and nutrition to local markets

Action plan

De-duplicated next steps, most important first.

#	Action plan	Area
1	Coverage is below 30% of the core metric set. Start with: Payments to Supplier Individuals: Smallholder, Water Withdrawn, Waste Produced: Total.	Metrics

#	Action plan	Area
2	Measure HOW MUCH: track scale (# stakeholders), depth (degree of change), and duration	Impact
3	Assess CONTRIBUTION: establish counterfactual or benchmark comparison	Impact
4	Evaluate RISK: assess evidence risk, execution risk, and external risk factors	Impact
5	Map each SDG claim to at least one IRIS+ metric with reported data	Evidence quality
6	Report the negative impacts that matter for agriculture, e.g. Greenhouse Gas Emissions: Direct (Scope 1) (OI4112); Land Directly Controlled: Treated with Pesticides (OI2569); Pesticide Hazard Classification Type (OI7394)	Evidence quality
7	Include risk-oriented and negative-impact metrics alongside positive outcomes	Evidence quality
8	Report Energy Savings from Services Sold (PD4927); Energy Consumption of Product Replaced (PD5578); Energy Generated for Sale: Non-Renewable (PI2210) to strengthen SDG 7 evidence (2/133 currently tracked)	SDGs
9	Add impact themes: Diversity And Inclusion, Agriculture	SDGs
10	Add impact themes: Climate, Climate Change Mitigation	SDGs

Methodology and definitions

How this report was produced

- Claims are extracted from the documents and graded on the NESTA standards of evidence; quantities are mapped to IRIS+ metric IDs only when unit and context are unambiguous.
- Five Dimensions scores (1–5) come from reported metrics where available and from document text otherwise; the evidence label says which.
- SDG scores (0–100) combine goal-specific metric coverage, inference from the business and impact themes; cross-cutting metrics count at reduced weight.
- Greenwashing risk weights claim–metric gaps (30%), missing negative impacts (20%), vague language (20%), selective reporting (15%) and lack of verification (15%).
- The IC gate applies the fund thesis thresholds; failures caused by missing data are reported as insufficient evidence, not as a negative finding.

- Scores use methodology v1.2.0 (config e0d5e058eba0); reports are comparable only when both match.
- Expected impact (methodology 2.2.0, illustrative): reach × depth × duration × (1 − deadweight) × attribution × probability of success; each factor's spread follows its NESTA evidence level; 4000 seeded Monte Carlo draws give P10/P50/P90. Evidence quality and data completeness are reported separately, never multiplied in.

AI & automation disclosure

Automated analysis: produced by rules and a deterministic scoring model; no generative AI was used. 3 of 22 scores are estimated from document text rather than reported metrics. Review before relying on it.

Claim extraction	rules
Score calculation	deterministic model
Metric / SDG tagging	rules
Narrative drafting	not used

Disclosed under the EU AI Act Art 50 transparency rules.

Terms used in this report

Impact claim
: A statement in a pitch deck or report about the change a company creates ("we connected 42,000 households"). Impact Vision extracts each claim, maps it to a metric where possible and grades how well it is evidenced.

IRIS+
: The Global Impact Investing Network's catalogue of standard impact metrics (about 790 in version 5.3c), each with a short ID such as OI2764 for "Greenhouse Gas Emissions Avoided". Using the same IDs makes results comparable across companies and funds.

Core metric set
: The short list of IRIS+ metrics a company in a given sector should report. Coverage is the share of that list the company reports; Impact Vision picks the list by sector.

Five Dimensions of Impact (5D)
: The Impact Management Project's framework for describing impact: What outcome occurs, Who experiences it, How Much change (scale, depth, duration), the investor's and company's Contribution compared with what would have happened anyway, and the Risk that impact doesn't happen. Impact Vision scores each dimension from 1 to 5.

Evidence-based / partial / estimated
: How a score was derived. Evidence-based scores rest on several reported metrics; partial scores on a few; estimated scores only on the text of the documents. Treat estimated scores as a starting hypothesis, not a finding.

SDG alignment

How strongly the company's activities and metrics relate to each of the 17 UN Sustainable Development Goals (0-100). A goal is "material" when the company claims it, its business clearly relates to it, or it reports a goal-specific metric; other goals are context only.

NESTA standards of evidence
: A five-level scale for how strong evidence of impact is: 1 = a narrative, 2 = data showing change, 3 = a comparison or control group suggesting the company caused it, 4-5 = independent or replicated rigorous evaluations.

Third-party verification
: An independent organisation (auditor, certifier, regulator) has checked a figure or practice. It raises confidence in a claim and lowers greenwashing risk.

Additionality
: Whether the outcome would have happened anyway without the company or investor. Strong additionality means the impact depends on them.

Greenwashing (impact-washing) risk
: A 0-100 score of how far a company's claims run ahead of its evidence: vague language, claims without metrics, missing negative impacts, selective reporting and no verification. Below 60 it mostly reflects missing data; at 60 or above it is a finding in itself.

IC gate
: The fund's minimum thresholds before a deal goes to the investment committee (5D score, top SDG score, DD coverage, greenwashing risk, exclusions). "Insufficient evidence" means the failures are missing data; "Fail" means a negative finding.

Scope 1, 2 and 3 emissions
: Greenhouse-gas emissions from a company's own operations (Scope 1), from the energy it buys (Scope 2) and from its value chain (Scope 3), as defined by the GHG Protocol.

tCO2e
: Tonnes of carbon-dioxide equivalent: a common unit that converts all greenhouse gases to the warming effect of CO2.

ESRS / CSRD / VSME
: The EU sustainability reporting rules (CSRD) and their standards (ESRS). The revised ESRS became law in September 2026 and apply from 2027. VSME is the voluntary standard for smaller companies; from 2027 large reporters may not ask partners with up to 1,000 employees for more than VSME data.

AI & automation disclosure

Methodology v2.2.0 (expected impact, IC gate) · v1.2.0 (5D, SDG, greenwashing) · config e0d5e058eba0

Generated by Impact Vision — open-source impact measurement.